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General Comment

See attached file(s)

Attachments

Encode OSTP RFI AI Action Plan Comment

March 14, 2025

Via Electronic Submission

Faisal D'Souza, NCO

Office of Science and Technology Policy

Executive Office of the President

2415 Eisenhower Avenue

Alexandria, VA 22314

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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1. Advancing Science and Innovation

1.1 AI Reliability Research at DARPA

The Department of Defense (DOD) faces significant challenges integrating advanced AI systems due to the limited reliability, explainability, and security of current technologies, which are often factors in high-stakes military decisions. To address these issues, DARPA is positioned to lead research in rigorous testing against adversarial threats, improved model interpretability, and secure hardware design. Investing in these areas will enable the DOD to assess AI systems effectively, ensuring their reliability before deployment in mission-critical situations. Consequently, increased investment in AI reliability research should be prioritized to maintain U.S. strategic advantage in national security.

1.2 Establish and Fund the National AI Research Resource (NAIRR)

Research at American universities laid the groundwork for modern artificial intelligence. It was professors at Stanford and Carnegie Mellon University that developed the algorithms for neural networks and deep learning that created the foundation for today's advances in artificial intelligence.[1] In the 1960s and 1970s researchers at UC Berkeley and Stanford made revolutionary advances in semiconductors and microprocessors that paved the way for Silicon Valley itself.[2] ARPANET, the foundational precursor to today's internet, was a joint project at UCLA, Stanford, and MIT funded by DARPA.[3] Without well-funded research at top institutions, America would lag far behind our competitors.

Today, a lack of computational resources and access to critical data is stifling innovation at American academic institutions. The National AI Research Resource (NAIRR) would patch that gap. Through collaborations with agencies, universities, and industry, it has already supported over 250 research projects and served institutions across the country.[4] Permanently establishing and funding NAIRR would provide American universities with the computational resources, data, software, and training they need to keep making breakthroughs.

Building on the foundation that NAIRR provides to academic institutions, America needs a comprehensive national strategy for scientific advancement through AI. Such a strategy would create what could be described as a "superhighway for science" - connecting our universities' research capabilities with national laboratories and industry partners in a cohesive ecosystem. This approach would integrate academic innovation with advanced computing resources, scientific datasets, and domain-specific AI models to tackle grand challenges in energy, materials science, and healthcare. By enabling researchers to move beyond computational constraints and fostering multidisciplinary collaboration, we can dramatically reduce the time from basic research to practical application. This acceleration would not only maintain America's scientific leadership but would transform how we address urgent national priorities, shifting breakthrough timelines from decades to years or even months.

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2. Strengthening Our Defenses

2.1 Biorisk, WMD Policy, and Gene Synthesis Screening

Advanced AI has already helped us solve some of the biggest unsolved problems in science (like discovering how proteins fold), but it might also empower malicious actors to inflict large-scale harm. One of the clearest areas of potential risk is AI being misused to aid in the creation or novel design of dangerous biological agents. We know that terrorist organizations

[1] Stanford University School of Engineering, "Laying the Foundation for Today's Generative AI," accessed March 15, 2025, https://engineering.stanford.edu/news/laying-foundation-todays-generative-ai.
[2] UC Berkeley Library, "The Berkeley Remix Podcast: Season 4, Episode 2: Berkeley Lightning: A Public University's Role in the Rise of Silicon Valley," July 24, 2019, accessed March 15, 2025, https://update.lib.berkeley.edu/2019/07/24/the-berkeley-remix-podcast-season-4-episode-2-berkeley-lightning-a-public-universitys-role-in-the-rise-of-silicon-valley/.
[3] Britannica, "ARPANET," accessed March 15, 2025, https://www.britannica.com/topic/ARPANET.
[4] National Artificial Intelligence Research Resource. "National Artificial Intelligence Research Resource (NAIRR)." Accessed March 15, 2025. https://nairrpilot.org/.

like al-Qaeda spent years trying to develop biological weapons, but lacked the technical sophistication to succeed.[5] OpenAI, for example, recently warned that its models are “on the cusp of being able to meaningfully help novices create known biological threats.”[6]

One clear way to mitigate this risk is requiring gene synthesis companies to screen for dangerous biological agents. Some government agency contracts currently enforce certain standards for purchasing synthetic nucleic acids and benchtop DNA synthesis equipment, and these standards should be maintained and expanded. This includes incorporating new sequences of concern derived from AI-driven research and collaborating with agencies such as HHS, DARPA, and the AI Safety Institute (AISI) to maintain real-time databases of emerging threats. Over the long term, the administration should work with international partners to promote universal screening standards, making it far more difficult for malicious actors to acquire or engineer dangerous pathogens. By taking a proactive stance on AI-fueled biorisk, the United States can help avert significant threats to global health and security.

The federal government's purchasing power represents an untapped tool for encouraging responsible AI development with a specific focus on preventing weapons of mass destruction. Instead of creating new regulatory frameworks, procurement requirements could establish market incentives for AI companies to implement rigorous safeguards against their systems being used to develop biological, chemical, nuclear, or radiological weapons. This approach would formalize practices many leading AI companies already claim to follow, but in a consistent and verifiable way aligned with established legal definitions of WMDs. By making government contracts contingent on having robust prevention policies, this light-touch strategy would affect only the most advanced AI systems while providing companies with regulatory certainty.

2.2 Frontier AI as Critical Infrastructure (Cyber Defense)

As AI systems grow in complexity and criticality, large-scale "frontier AI" models increasingly resemble essential infrastructure across finance, healthcare, logistics, and defense. These systems process vast amounts of sensitive data and power critical decision-making tools, yet present high-value targets for cyber threats. Currently, gaps exist in defining frontier AI as critical infrastructure, creating ambiguity around mandatory reporting and government's role in incident response.

Frontier AI introduces unique cyber security challenges through expanded attack surfaces, vulnerabilities specific to machine learning architectures, and potential cascading failures when compromised systems affect multiple sectors. These risks are fundamentally different from traditional IT security concerns and require specialized approaches to mitigation.

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[5] Mowatt-Larssen, Rolf. Al Qaeda Weapons of Mass Destruction Threat: Hype or Reality? Cambridge, MA: Harvard Kennedy School, Belfer Center for Science and International Affairs, 2010.
[6] OpenAI. "Deep Research System Card." Accessed March 15, 2025. https://openai.com/index/deep-research-system-card/.

The federal government should direct NIST to develop cyber security frameworks tailored to frontier AI systems. These frameworks should address model security, legacy software integration, and resilience against AI-specific attacks in collaboration with international partners. Formally designating frontier AI companies and their data centers as a critical infrastructure sector would bring them under the Cyber Incident Reporting for Critical Infrastructure Act (CIRCIA), requiring prompt disclosure of significant breaches to federal cyber security agencies.

This designation should be accompanied by expanded capabilities at DHS and other relevant agencies. This includes specialized personnel and technologies focused on AI cyber security, dedicated AI threat response teams, and advanced monitoring tools. Additionally, investment in open-source security testing tools for AI systems would democratize access to security capabilities for smaller organizations while fostering a broader community of security researchers.

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3. Visibility and Understanding

3.1 Whistleblowers and National Security

Employees embedded in AI labs are uniquely poised to discover emerging vulnerabilities that can pose threats to U.S. national security. Currently, companies have no obligation to provide employees a way to report those threats internally and can retaliate against employees who alert the government. Existing whistleblower protections do not cover many of the risks that an employee may want to report. This lack of legal protection could leave the U.S. vulnerable to serious risks that employees may otherwise seek to alert the government of. With few other paths for agencies to be alerted to or become aware of serious threats it is critical that we protect the workers most aware of the state of this technology from retaliation.

Multiple steps can be taken to facilitate narrow reporting of serious national security risks. Agencies such as the Department of Commerce (DOC) or the Department of Defense (DOD) should establish secure channels for confidential disclosures about AI-related concerns. Those channels should be monitored by experts on the issue that can understand the level of risk and escalate to agency leadership when necessary. Portals like this would grant anonymity and make it clear to employees that their report will be heard by someone who understands the issue. This would reflect longstanding practices in other high-security domains like national intelligence, the judicial system, and financial services.

In tandem, Congress should clarify and expand whistleblower statutes to include AI-driven risks, ensuring that employees can safely, securely, and anonymously report serious risks to the correct government actors. These protections should focus on substantial and specific risks to national security. Such protections would mirror critical industries like nuclear energy, oil and gas, and transportation infrastructure where employees are protected when reporting technically legal but dangerous conduct. These protections would increase the government's understanding of the evolution of AI and its risks without adding new burdens on the companies developing this critical technology.

3.2 Supporting Standards Development and Evaluation

Without the ability to evaluate and understand the state of frontier AI development the U.S. will inevitably fail to recognize critical risks and fall behind our adversaries. The AI Safety Institute (AISI) is the best repository of technical knowledge and capabilities that the U.S. currently has. It has already successfully worked with the top companies to evaluate the state of frontier AI and developed standards for AI that are our best hope of influencing international AI development. The technical expertise at AISI remains our best tool to assess foreign models like DeepSeek R1 and track our progress in comparison to our adversaries. As new risks emerge from frontier models AISI is the best positioned body to keep agency leadership informed and ensure our national security apparatus is never caught off guard. It is critical not only that the work of AISI continues but that its funding and staff is maintained and expanded.

The impact of such a body goes far beyond the United States itself. As oppositional countries like China seek to influence the future of AI internationally, bodies like AISI under NIST represent our strongest opportunity to encode American values into the future of this technology. China is already aggressively working to influence international standards bodies, flooding groups like the ITU with hundreds of new standards designed to benefit Chinese companies and embed authoritarian values into the technologies of the future. China views standards setting as one of the most important factors for the economic future of China and has already succeeded in influencing technical standards on technologies like 6G mobile.[7] The Chinese government directly subsidizes domestic firms' participation in standards bodies, rewarding both the proposal and successful setting of new standards.[8] University professors and researchers are similarly rewarded and Chinese academic institutions are actively recruiting foreign experts in science and technology to take part in their standards-setting work through the "Recruitment Program of Global Experts."[9] Meanwhile, the National Institute of Standards and Technology has publicly stated that its work on developing AI standards would be "very, very tough" due to a serious lack of funding.[10]

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[7] "China Is Writing the World's Technology Rules," The Economist, October 10, 2024, https://www.economist.com/business/2024/10/10/china-is-writing-the-worlds-technology-rules.
[8] United States-China Business Council. China in International Standards Setting: USCBC Recommendations for Constructive Participation. February 2020. https://www.uschina.org/wp-content/uploads/2020/02/china_in_international_standards_setting.pdf.
[9] "University professors and researchers are similarly rewarded and Chinese academic institutions are actively recruiting foreign experts in science and technology to take part in their standards-setting work through the 'Recruitment Program of Global Experts.'"
[10] Kelley, Alexandra. "NIST's Emerging Tech Work Will Be 'Very Difficult' Without Sustained Funding, Director Says." Nextgov, May 22, 2024.

To compete with China internationally and maintain our influence on advanced technology abroad we must take our role in standards setting seriously. We can support NIST's work by working with Congress to increase its funding through both appropriations and private donations. The administration could pursue a new Foundation for Standards and Metrology, similar to the body proposed by the Expanding Partnerships for Innovation and Competitiveness Act modeled on existing foundations that support other federal agencies like the CDC and NIH. One key barrier to standards meetings being hosted in America is substantial visa backlogs and restrictions. A program similar to the NSA National Centers for Academic Excellence could be established to recognize schools that conduct outstanding work in standards development to channel new funding and support their work. Cooperation with American allies is another way to counter China's international influence. By including standards activities in bilateral and multilateral science and technology cooperation agreements, the U.S. could leverage its existing influence to counter China's.

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4. Infrastructure/Data Centers

4.1 Key Provisions from Prior Data Center EO

The January 2025 executive order took initial steps by making federal lands available for AI development, but failed to address permitting bottlenecks and imposed restrictive clean energy mandates. We must build upon this foundation by leveraging Department of Defense and Energy sites for data centers with truly expedited NEPA reviews through categorical exclusions. While China commits nearly $140 billion to state-directed AI infrastructure, American companies need a streamlined regulatory pathway that maintains necessary oversight without years of bureaucratic delay.[11]

4.2 Grid Connections and Energy Sources

High-performance data centers require immediate access to firm, uninterrupted power at unprecedented scale. To meet this demand, we must fast-track the recommissioning of existing nuclear plants specifically for AI facilities, following Microsoft's successful Three Mile Island model. Simultaneously, we should deploy enhanced geothermal systems and small modular reactors by establishing federal loan guarantees and regulatory fast-tracks with specific deployment timelines. During this transition, the Defense Production Act should be leveraged to

https://www.nextgov.com/emerging-tech/2024/05/nists-emerging-tech-work-will-be-very-difficult-without-sustained-funding-director-says/396803/.

[11] "China Steps Up Support for Tech Sector as AI Enthusiasm Soars." Bloomberg, March 6, 2025. https://www.bloomberg.com/news/articles/2025-03-06/china-steps-up-support-for-tech-sector-as-ai-enthusiasm-soars.

resolve supply chain bottlenecks for gas turbines and other critical components needed for transitional power generation.

The administration should establish "Special Compute Zones" on federal lands with categorical NEPA exclusions, pre-approved grid connections, and expedited 30-day permitting timelines. Companies accessing these benefits must implement specific security protections, including DOD's Cybersecurity Maturity Model Certification (Level 3), background screening for personnel accessing sensitive hardware, and supply chain security measures with intelligence community oversight. By coupling energy access with security requirements, we ensure that as we build American AI capabilities, we simultaneously protect them from foreign adversaries without imposing arbitrary restrictions on how these facilities are powered.

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5. Chips/Export Controls

5.1 Furthering the Framework for Artificial Intelligence Diffusion

Export controls play a pivotal role in safeguarding cutting-edge U.S. AI capabilities from potential adversaries. The Framework for Artificial Intelligence Diffusion is a sensible continuation of technology export controls that began in the first Trump Administration with their work on countering Huawei and ZTE. The framework may need to be modified to reflect changing technical and geopolitical realities, but broadly continuing it gives the US a strong path to incentivize other nations to implement basic KYC and security policies to prevent the capture or misuse of this tremendously consequential technology.

5.2 Funding BIS and CHIPS Act Implementation

The Bureau of Industry and Security (BIS) bears an expanding mandate in overseeing AI hardware export controls, yet suffers from limited budgets and outdated IT infrastructure.[12] Strengthening BIS's enforcement capacity through sustained funding—such as an additional $75 million annually for personnel and a one-time $100 million upgrade for monitoring systems—would enable more rigorous oversight of emerging AI technologies. These resources would facilitate deeper technical expertise, faster case processing, and stronger international investigations into potential violations.

While we work to hinder China's access to our technology, we must also bolster domestic production. Domestic semiconductor production has gained momentum, but maintaining this trajectory requires continued investment in research and development as well as enhanced

[12] Graham, Edward. "Export Control Agency Lacks Funding and Tech to Manage Its Workload, Official Says." Nextgov, March 21, 2024. https://www.nextgov.com/digital-government/2024/03/export-control-agency-lacks-funding-and-tech-manage-its-workload-official-says/395132/.

supply chain resilience. Expanding tax incentives, supporting specialized chip design facilities, and strengthening stable partnerships with allied nations for securing raw materials are critical steps toward advancing U.S. semiconductor manufacturing. By bolstering regulatory support and effectively implementing existing policy initiatives, the United States can sustain its global leadership in AI-specific hardware, a cornerstone of economic competitiveness and national security.

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6. Talent

6.1 Immigration Reforms for AI Expertise

America's leadership in AI has long benefited from foreign-born talent attracted by the country's research ecosystem and innovation culture. One study found that of the 50 "most promising" U.S.-based AI startups on Forbes' 2019 "AI 50" list, 66% had an immigrant founder.[13] The most common path for immigrants to found U.S.-based startups is through student visas, but current restrictions force many foreign students to stay locked into academia. Among those who stay on that path, many AI PhDs are forced to leave, and one-third state that immigration constraints played a key role. Losing such talent at a time when skilled AI scientists are in dramatic demand and stunningly low supply is a mistake.

A number of easy changes could be made to attract foreign talent and keep it in the United States. Creating a subcategory for H-1Bs for AI workers with a higher annual cap and extended baseline duration would allow American companies to bring in more skilled AI scientists and provide a clear path for them to stay in the United States and contribute to American innovation. The O-1 visa's dual-intent classification is currently unclear, and many startup founders have had applications rejected. Making it explicit that the O-1A is a status with unequivocal dual intent status would clarify that issue. The administration could also recapture some of the over 300,000 green cards that go unused and provide them to companies seeking to attract skilled AI workers.[14] Another source of untapped pathways for AI workers is DoD's H-1B2 program. DoD currently only uses ~30% of its 100 allotted visas. The remaining 70% could be easily allocated in the direction of AI projects at DoD. The Schedule A shortage occupation list, a set of occupations that get streamlined labor certification processes, is rarely updated and doesn't include critical STEM occupations where there are large scale shortages like AI. Updating it would create new employment-based green cards and streamline typically slow labor

[13] "University professors and researchers are similarly rewarded and Chinese academic institutions are actively recruiting foreign experts in science and technology to take part in their standards-setting work through the 'Recruitment Program of Global Experts.'"
[14] Esterline, Cecilia. "Green Card Backlogs Block Billions in Rural Investments." Niskanen Center, June 6, 2024.
https://www.niskanencenter.org/green-card-backlogs-block-billions-in-rural-investments/.:contentReference[oaicite:0]{index=0}

certification processes. Finally, permanent labor certification via special handling could be granted to employers seeking AI talent. Currently only universities making teaching hires are granted special handling. Expanding this definition to employers hiring AI talent could substantially streamline hiring in cases where American companies could benefit from advanced foreign talent. Such an expansion could be accomplished by allowing the National Science Foundation to occasionally add key emerging technologies to a list that qualifies for National Interest Waiver EB-2 I-140 Petitions.

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7. AI and Protecting Children

7.1 Deepfake Porn and Non-Consensual Intimate Imagery

Surveys today indicate that nearly one in five high schoolers say they know of a peer at their school who's been victimized by AI-generated nude images.[15] That statistic would have been unfathomable just a few years ago when it took hundreds or thousands of images to create a realistic deepfake image of someone. Now all anyone needs is a single, clothed image of a victim to create realistic nude imagery. The resulting flood of AI-generated nudes in American schools is unignorable.

To protect our children we need new laws, companies cooperating with the federal government, and updated policies in our schools. The President and First Lady's outspoken support for the TAKE IT DOWN Act is a critical step towards accountability for perpetrators and support for victims. By criminalizing authentic and deepfake non-consensual intimate imagery and requiring social media companies to remove images reported by victims, we can prevent this crisis from spiraling out of control. The DEFIANCE Act, another bill that passed the Senate unanimously last session, would add a second layer of support by creating a civil remedy for victims, allowing them to pursue injunctive relief to end the distribution of images and receive damages for harm caused to them.[16]

Beyond legislation we need companies to work with the Trump Administration to restrict the creation and distribution of AI-generated nudes. Payment service companies like Cash App and Square should agree to bar deepfake porn websites from using their services, and invest in the broader detection and mitigation of payments for the creation or procurement of image-based

[15] Center for Democracy & Technology, "In Deep Trouble: Surfacing Tech-powered Sexual Harassment in K-12 Schools," accessed March 15, 2025, https://cdt.org/insights/report-in-deep-trouble-surfacing-tech-powered-sexual-harassment-in-k-12-schools/.

[16] U.S. Congress. Senate. S. 3696 – Disrupt Explicit Forged Images and Non-Consensual Edits Act of 2024. 118th Cong., 2nd sess., 2024. https://www.congress.gov/bill/118th-congress/senate-bill/3696.:contentReference[oaicite:0]{index=0}

sexual abuse. We need companies like Google, Meta, and Bing to restrict the appearance of image-based sexual abuse in search results and on platforms. Companies that are developing image-generation models must agree to filter the datasets used to train their models to remove image-based sexual abuse and child sexual abuse material. NIST can work to support companies by investing in sophisticated deepfake detection and watermarking tools so platforms have additional tools to identify and remove harmful material online.

School districts should explicitly ban the creation and circulation of deepfake sexual imagery in their policies, communicate these rules widely through orientations and events, and treat fabricated intimate images with the same seriousness as authentic ones. Clear and consistent consequences for perpetrators, potentially including suspension or expulsion, should be established, and school officials must promptly inform law enforcement or child protective services when required. In parallel, administrators should adopt standardized procedures to protect victims' identities and ensure they receive confidential support, including training staff on how to handle deepfake incidents. By incorporating safeguards into existing codes of conduct, sexual harassment, and cyberbullying policies, schools can deter would-be perpetrators, respond quickly and appropriately when abuse occurs, and better safeguard students' well-being.

7.2 Emotional Chatbots

AI Chatbots designed to supplant human relationships are a new frontier of harm for America's youth. Young people, particularly those who are already lonely or struggling, have real potential to become dependent on these applications for companionship and validation. Childhood is meant to be a time for young people to struggle through complex relationships in safe environments free from exploitation. If they are able to avoid the difficulties of building those relationships by conversing with artificial entities their ability to be well adapted adults will be seriously damaged. Already we are seeing young people that spend large portions of their days conversing with these artificial companions, and in extreme cases resorting to self harm or suicide.[17] We may be experiencing a situation similar to the early days of social media, when we missed our chance to intervene and establish common-sense guardrails for children. We urge the administration to proactively monitor this particular application of AI for manipulative design practices and deceptive marketing and, if necessary, intervene to protect our children.

Conclusion

The above recommendations offer a strategic approach to sustaining America's leadership in artificial intelligence. By bolstering foundational R&D, strengthening defenses against national

[17] Tiffany Hsu and Sapna Maheshwari, "AI Chatbots Are Being Blamed in a Teen's Suicide. Can Companies Be Held Liable?," New York Times, October 23, 2024, https://www.nytimes.com/2024/10/23/technology/characterai-lawsuit-teen-suicide.html.

security threats, ensuring adequate oversight and whistleblower protections, expanding infrastructure and data center capacity, refining export controls, investing in a robust AI talent pipeline, and immediately tackling harmful applications like deepfake pornography, the United States can chart a future that safely harnesses AI’s transformative potential.
